

```

import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset from the CSV file
df = pd.read_csv('titanic.csv')

# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2, figsize=(12, 12))

# write the code..

count=df['Pclass'].value_counts().sort_index()
count_gen=df['Gender'].value_counts()
count_s=df['Survived'].value_counts().sort_index()

axes[0,0].bar(count.index,count.values,color=['skyblue'])
axes[0,0].set_title('Passenger Class Distribution')
axes[0,0].set_xlabel('Pclass')
axes[0,0].set_ylabel('Count')

axes[0,1].pie(count_gen.values,labels=['male','female'],colors=
['lightblue','lightcoral'],autopct='%1.1f%%')
axes [0,1].set_title('Gender Distribution')

axes[1,0].hist(df['Age'],color=['lightgreen'],edgecolor='black',bins=8)
axes [1,0].set_title('Age Distribution')
axes[1,0].set_xlabel('Age')
axes[1,0].set_ylabel('Frequency')

axes[1,1].bar(count_s.index,count_s.values,color=['lightblue','lightcoral'])
axes [1,1].set_title('Survival Count')
axes[1,1].set_xlabel('Survived (0 = No, 1 = Yes)')
axes[1,1].set_ylabel('Count')

axes[2,1].plot()
axes[2,1].set_xlim(0,1)
axes[2,1].set_ylim(0,1)

```

```
import pandas as pd
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# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0],
inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)

# Write your code here for Histogram
plt.hist(data['Age'], bins=30, edgecolor='black')
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.title("Age Distribution")

plt.show()
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data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot f

data["Survived"].value_counts().plot(kind="bar")
plt.title("Survival Count")
plt.xlabel("Survived")
plt.ylabel("Count")

plt.show()
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# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Gender

grouped = pd.crosstab(data['Sex'], data['Survived'])
# grouped = data.groupby(['Sex', 'Survived']).size().unstack()
grouped.plot(kind='bar', stacked=True)
plt.xlabel('Gender')
plt.ylabel('Count')
plt.title('Survival by Gender')
plt.legend(['Not Survived', 'Survived'])
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data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Pclass

grouped = pd.crosstab(data['Pclass'], data['Survived'])
grouped.plot(kind='bar', stacked=True)

plt.xlabel('Pclass')
plt.ylabel('Count')
plt.title('Survival by Pclass')
plt.legend(['Not Survived', 'Survived'])

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data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Embarked
grouped = pd.crosstab(data['Embarked_Q'], data['Survived'])
grouped.plot(kind='bar', stacked=True)

plt.xlabel('Embarked')
plt.ylabel('Count')
plt.title('Survival by Embarked')
plt.legend(['Not Survived', 'Survived'])

plt.show()
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data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Age by Pclass

plt.figure(figsize=(8, 6))
data.boxplot(column='Age', by='Pclass')

# Plot customizations
plt.title('Age by Pclass')
plt.suptitle('') # Remove the default subtitle
plt.xlabel('Pclass')
plt.ylabel('Age')

# Show the plot
plt.show()
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data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Age by Survived
plt.figure(figsize=(8, 6))
data.boxplot(column='Age', by='Survived')

# Plot customizations
plt.title('Age by Survival')
plt.suptitle('') # Remove default subtitle
plt.xlabel('Survived')
plt.ylabel('Age')

# Show the plot
plt.show()
```



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data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Fare by Pclass
plt.figure(figsize=(8, 6))
data.boxplot(column='Fare', by='Pclass')
plt.title('Fare by Pclass')
plt.suptitle('') # Remove the default subtitle
plt.xlabel('Pclass')
plt.ylabel('Fare')
# plt.grid(False)
plt.show()
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data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Fare by Pclass

plt.figure()
plt.scatter(data['Age'], data['Fare'])
plt.title('Age vs. Fare')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
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data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Scatter Plot for Age vs. Fare by Survived

colors = data['Survived'].map({0:'red',1:'blue'})
plt.scatter(data['Age'],data['Fare'],c=colors)
plt.title('Age vs. Fare by Survival')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
```